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[22CSBS215]

FIRST SEMESTER B.Tech EXAMINATION, DECEMBER 2022

DISCRETE MATHEMATICS

Time: 3 Hours **Maximum Marks: 100**

Instructions: Missing data may be suitably assumed.

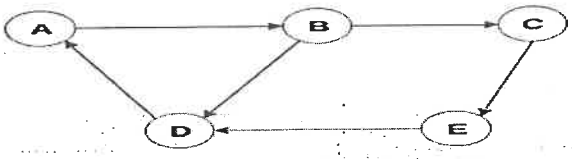
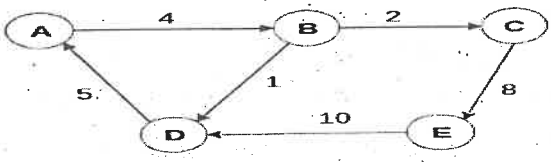
PART-A (5 X 2=10 Marks)

ANSWER ALL QUESTIONS

1.	**a.**	Define conjunction and write down its truth table.	L1	CO1	**02**
	b.	Demonstrate the generating function for the sequence 1,2,3,4,...	L2	CO2	**02**
	c.	Compute how many distinguishable strings of length 3 over the alphabet {A,B, . . . ,Z} exist?.	L3	CO3	**02**
	d.	If $A = \{1,2,3\}$ & $B = \{2,3,4\}$ then determine $\lvert A - B\rvert$, $\lvert B - A\rvert$, $\lvert A \cap B\rvert$, and $\lvert A \cup B\rvert$.	L3	CO4	**02**
	e.	Define OR Logic Gate.	L1	CO5	**02**

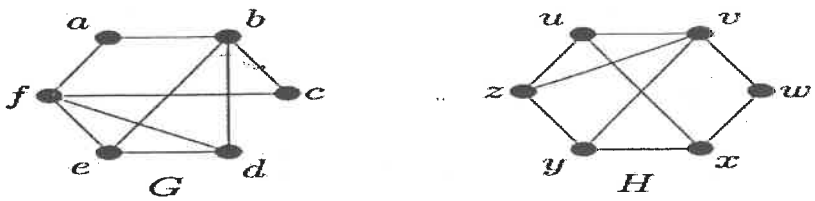
PART-B (6X5=30)

2.	**a.**	Prove the logical equivalence $(\neg p \wedge q) \vee \neg (p \vee q) \Leftrightarrow \neg p$.	L3	CO1	**05**
		OR			
	b.	Show that $(p \vee q) \rightarrow r)$ is logically derived from $p \rightarrow r$ & $q \rightarrow r$.	L3	CO1	**05**
3.	**a.**	There are 50 baskets of apples. Each basket contains no more than 24 apples. Show that there are at least 3 baskets containing the same number of apples.	L3	CO2	**05**
		OR			
	b.	Solve the recurrence relation $a_{n+1} = 4a_n$ given that $a_0 = 3$.	L3	CO2	**05**
4.	**a.**	Use appropriate counting principle to find how many different license plates can be generated, if each plate contains a sequence of three letters followed by 3 digits?	L3	CO3	**05**
		OR			
	b.	The number of virus affected files in a system is 1000 [to start with] and it increases 250% every two hours. Use recurrence relation to determine the number of virus affected files in the system after one day.	L3	CO3	**05**
5.	**a)**	Let $A = \{2,4,6,8\}$ and $B = \{1,2,3\}$ and the relations R_1, R_2, R_3, R_4 from A to B, which are defined as follows : **i.** aR_1b if $a \leq b$ **ii.** aR_2b if $a > b$ **iii.** aR_3b if a *divides* b **iv.** aR_4b if b *divides* a	L3	CO4	**05**
		OR			

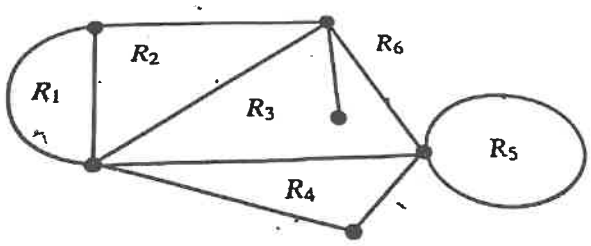
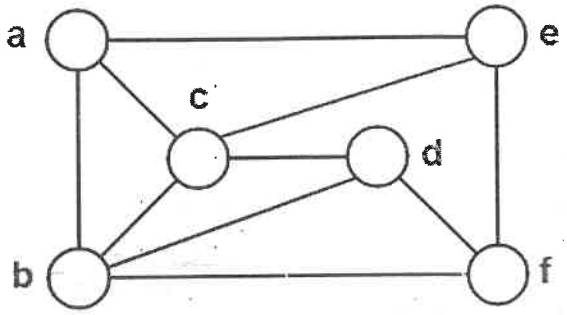
	b.	Show that the intersection of two subgroups of a group G is again a subgroup of G.	L3	CO4	05
6.	a.	Sketch the K-maps for the following Boolean expressions i. i. $xy + \bar{x}y$ ii. $x\bar{y} + \bar{x}y$ ii. iii. $x\bar{y} + \bar{x}y + \bar{x}\,\bar{y}$	L3	CO5	05
		OR			
	b.	Express the Boolean function $F = x + yz$ as a sum of minterms.	L3	CO5	05
7.	a.	Compute how many vertices are needed to construct a graph with 6 edges in which each vertex is of degree 2?.	L3	CO6	05
		OR			
	b.	Construct the adjacency matrix for the following graphs: i.  Directed Graph ii.  Weighted Directed Graph	L3	CO6	05

PART-C (6X10=60)

8.	a.	Construct the truth table for the following compound proposition $\{(p \vee q) \wedge [(p \rightarrow r) \wedge (q \rightarrow r)]\} \rightarrow r$ and check whether the given proposition is valid, satisfiable, and unfalsifiable or not.	L3	CO1	10
		OR			
	b.	Obtain the PDNF and PCNF of $p \vee (\neg p \rightarrow (q \vee (\neg q \rightarrow r)))$,.	L3	CO1	10
9.	a.	In a survey of 260 college students, the following data were obtained 64 had take mathematics course 94 had take computer science course 58 had taken business course 28 had taken both maths and business course, 26 had taken both maths and computer science course , 22 had taken both computer science and business, 14 had taken all 3 type of course, then compute i. How many students were survey who had taken none of the 3-types of course? ii. How many had taken only computer science?	L3	CO2	10
		OR			

	b.	Using principle of mathematical induction, prove that $1+2+3+\cdots+n=\frac{n(n+1)}{2},\ \forall n \geq 1.$	L3	CO2	**10**
10	a.	i. Prove the following statement through direct proof. *"The product of two odd numbers is odd"* ii. If 13 people are assembled in a room, show that at least two of them must have their birthdays in the same month.	L3	CO3	**5+5**
		OR			
	b.	Solve the recurrence relation $F_{n+2} = F_{n+1} + F_n\ \forall n \geq 0$ given $F_0 = 0,\ F_1 = 1.$	L3	CO3	**10**
11.	a.	Define equivalence relation. Show that the following relation R defined on the set of all integers Z is an equivalence relation, $R = \{(x, y): x, y \in Z \text{ and } (x-y) \text{ is an even number}\}$.	L3	CO4	**10**
		OR			
	b.	Show that the set of all integers is an abelian group with respect to addition (+) operation.	L3	CO4	**10**
12.	a.	i. Show that the following distributive laws are valid through truth table. i. $x(y+z) = xy + xz$ ii. $x + (yz) = (x+y)(x+z)$	L3	CO5	**10**
		OR			
	b.	Construct the logic circuits that produce the following outputs: i. $(x+y)\bar{y}$ ii. $\bar{x}\ \overline{(y+\bar{z})}$ iii. $(x+y+z)\ (\bar{x}\,\bar{y}\,\bar{z})$	L3	CO5	**3+3 +4**
13.	a.	i. Verify whether below two graphs G and H are Isomorphic or not ?.  ii. A connected planar graph G has 9 vertices with degrees 2, 2, 3, 3, 3, 4, 5, 6, 6. Compute the number of regions of G.	L3	CO6	**5+5**
		OR			

	b.	**i.**	Verify Euler's formula for the below planar graph.	L3	CO6	**4+6**



ii. Define Chromatic number and compute the Chromatic number for the following Graph